AI ACADEMY · LEVEL 4 · GRADE 9 · WEEK 23 · STUDENT LESSON

Hold Out Test Data

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Responsible Plant Classifier

Before you start

Week 13: final labels must not change the chosen model.

Plan time to read, investigate and explain. Keep predictions separate from observations.

STUDENT LESSON · 1 OF 18

Your mission

> Start with the target: Freeze the full evaluation procedure before testing. Try the recall before opening notes.

Your learning target

I can **freeze** the full evaluation procedure before testing.

I can report a held-out result with counts and limitations.

I can recognize when test feedback has become development.

Remember before reading

Close your notes first. A rough first answer helps us choose the right starting point; it is not a score for your ability.

Without opening the old lesson, explain one key idea from Train a Regression Model. Give an example and a mistake that the idea helps you avoid.

Without opening the old lesson, explain one key idea from Mean, Median, and Baselines. Give an example and a mistake that the idea helps you avoid.

Without opening the old lesson, explain one key idea from Precision and Recall. Give an example and a mistake that the idea helps you avoid.

After trying, check the relevant lesson or ask your teacher. Change the part of your explanation that the **evidence** shows was wrong.

The final test set must remain untouched until model and threshold choices are finished, or it becomes part of **training** by decision.



Your target

LEARNING TARGET After seeing test errors, a team adjusts its threshold and reports the improved score on that same test. Rename the set's role and describe an honest final report if no new data exists.

Builder name and experiment date:

Readiness for the connected explanation

Why is validation not an untouched final test?

This week's end goal

By the end, I can explain hold out test data using a traced example and a stated limitation.

STUDENT LESSON · 2 OF 18

Notice the evidence

> Investigate the supplied Hold Out Test Data case. Separate observations, given facts and assumptions before choosing an explanation.

PICTURE ENTRY

Notice → Predict → Explain

Begin with visible **evidence**; do not explain more than the picture and information support.



Start with this puzzle

MYSTERY The team must use hold out test data to solve a model-building problem. What information do they have? What important **evidence** is still missing?

Name three visible details without interpreting them.

Predict the data, model, or evaluation decision being made.

Identify one hidden assumption that could change the result.

State what **evidence** would confirm or reject your **prediction**.

My **prediction** and supporting **evidence**:

A closer explanation

A final test is valuable because it did not guide model choices. **Freeze** the model, **training** pre**process**ing, threshold, metric, missing-**prediction** policy and test membership. Record the version before opening test labels.

A classroom result is **evidence** about a small chosen set, not approval for deployment.

Use fictional classroom data throughout this investigation. Keep the task, units, reference labels and data split visible as you reason.

Earlier weeks supplied the model-building **process**; this week examines final testing and honest **claim**s.

STUDENT LESSON · 3 OF 18

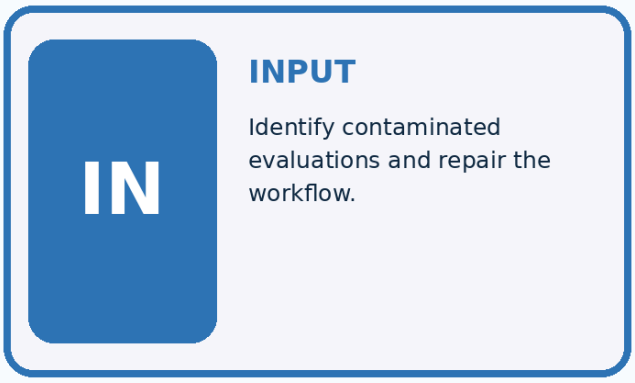
See how it works

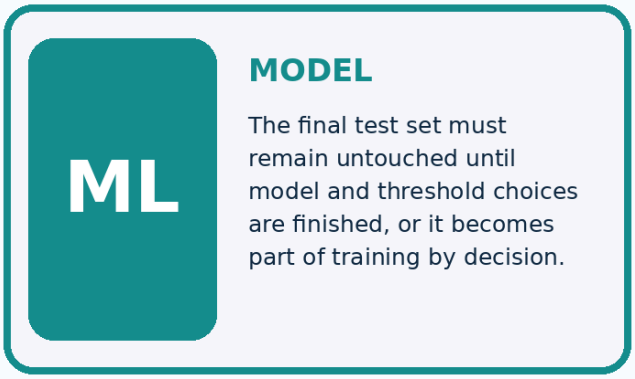
> Follow the mechanism for Hold Out Test Data. Connect each labelled input, deciding step and result.

ML MECHANISM

Input → Model → Evidence

Trace the information, fitted behavior, and evaluation proof separately.







Question to explain

QUESTION Identify contaminated evaluations and repair the workflow.

MODEL IDEA The final test set must remain untouched until model and threshold choices are finished, or it becomes part of **training** by decision.

PROOF NEEDED Use an unseen case, visible calculation, error analysis, and a stated limitation.

How the mechanism works

UNDER THE HOOD Team checks test accuracy after every parameter change. Those repeated choices adapt to the test set, contaminating it.

Draw a fresh data → model → **prediction** → evaluation trace:

A closer explanation

A frozen classifier is tested on 20 fictional cases: TN 12 FP 2 FN 3 TP 3, blocked positive. Accuracy is 15/20=75 percent, precision 3/5=60 percent and recall 3/6=50 percent.

Its previously fitted always-clear baseline gets 14/20=70 percent. The candidate improves accuracy here but misses half the blocked cases.

Before continuing, underline the decision rule and circle the **evidence** that supports the conclusion. Explain the example to a partner without replacing the counts or conditions with a stronger **claim**.

Keep the distinction between an observed result and a proposed next step.

Connect the picture and the explanation

Point to the **input**, the deciding step and the result in this week's visual. Explain the same step in ordinary words. A picture helps when you can say what each part represents.

STUDENT LESSON · 4 OF 18

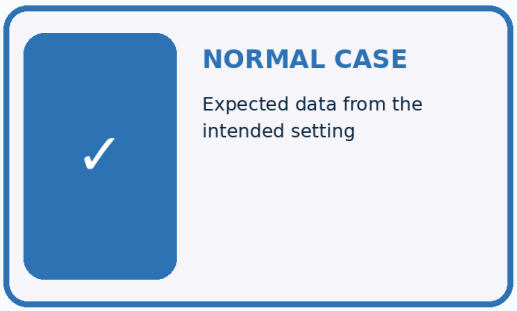
Compare the cases

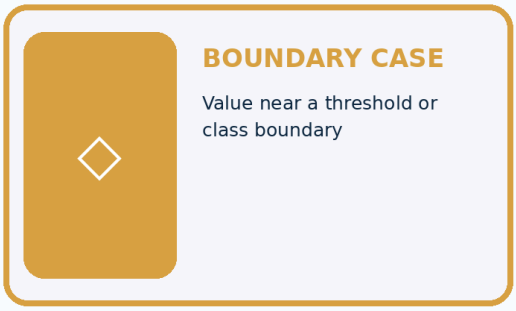
> Compare the supplied cases for Hold Out Test Data. Identify the detail that changes the conclusion.

EVALUATION DESIGN

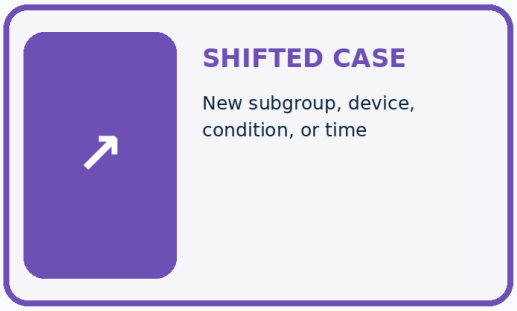
Four Tests, Four Kinds of Evidence

A system is not understood until it survives more than one comfortable example.









NORMAL Expected data and expected behavior.

Keep this boundary clear

BOUNDARY Value close to a threshold, ambiguous label, or uncertain **prediction**.

MISSING Necessary field, label, source, or measurement is unavailable.

SHIFTED New lighting, species, device, subgroup, or time condition.

RELEASE RULE Passing one case never proves generalization. Predefine expected behavior for all four categories.

Which case most threatens today's **claim**, and why?

Change one thing in the pictured case

Change one thing

Change: The team changes a threshold after inspecting the false negatives.

Predicted result

Prediction: This set is now development **evidence** for the revised model.

Why the result follows

Reason: Using final outcomes to change the model breaks the untouched hold-out role.

STUDENT LESSON · 5 OF 18

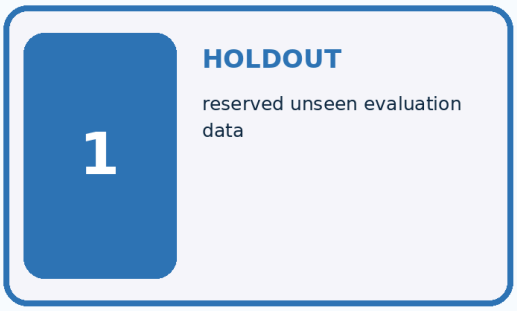
Words that unlock the idea

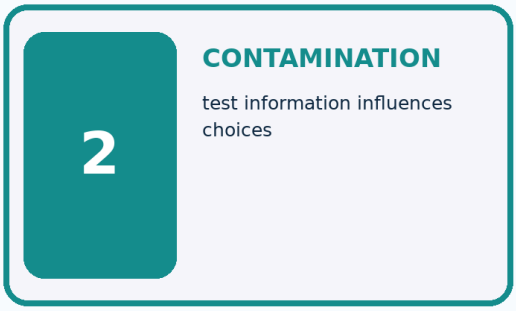
> Use the vocabulary of Hold Out Test Data to explain a concrete decision, rather than only memorising definitions.

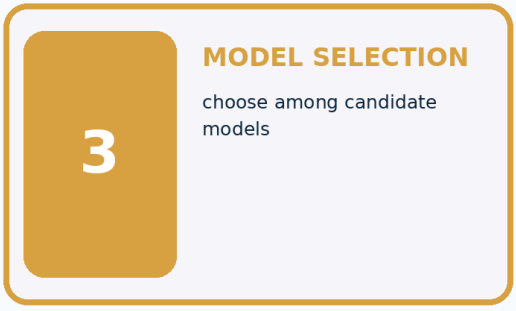
VOCABULARY

Words You Can Calculate and Defend

A definition matters when it lets you interpret a new dataset, model, or result.









Term	Precise meaning	My ML example
holdout	reserved unseen evaluation data	
contamination	test information influences choices	
model selection	choose among candidate models	
final estimate	performance measured once after choices freeze	

MISCONCEPTION TO REPAIR Test data remains unbiased as long as the model never directly fits its labels.

Repair the **claim** using a definition, calculation, or test record:

Words used in the closer explanation

Word	Meaning
Freeze	Record a fixed model and procedure before testing
Held out	Kept separate from development decisions
Provenance	A record of where data and results came from
Claim	A statement whose scope must match its evidence

STUDENT LESSON · 6 OF 18

Follow a worked example

> Follow the worked trace for Hold Out Test Data, predict a step, then compare with its explanation.

WORKED EVIDENCE

Reason Step by Step

Predict first. Then check each step. A correct answer without visible **evidence** cannot be audited.

WEEK 23 • WORKED EVIDENCE

1. Team checks test accuracy after every parameter change.
2. Those repeated choices adapt to the test set, contaminating it.
3. Use training for fitting and validation for model or threshold choices.
4. After freezing choices, open the holdout once and report the final estimate honestly.

Step 1

STEP 1 Team checks test accuracy after every parameter change.

Step 2

STEP 2 Those repeated choices adapt to the test set, contaminating it.

Step 3

STEP 3 Use **training** for **fitting** and validation for model or threshold choices.

Step 4

STEP 4 After freezing choices, open the **holdout** once and report the **final estimate** honestly.

CHECK Verify the denominator, unit, split, class definition, and data version before interpreting a score.

Recalculate, retrace, or restate the **evidence** independently:

A closer explanation

Final testing and honest claims at a glance

PART	EVIDENCE OR ACTION	INTERPRETATION
Before labels	Freeze version and rules	Record test membership
During evaluation	Keep every planned case	Count failures explicitly
After evaluation	Publish full results	Restrict the claim

The test comparison must use the same cases and consistent rules. Report the denominator, reference-label process and sampling limitations. If the software fails on one case, report that failure; do not silently drop the case and keep the original headline denominator.

Why does 75 percent versus 70 percent not settle deployment? Explain the steps or evidence behind your response.

Learn from the worked case

Cover the result before checking it. Say what is given, choose the procedure, and follow one step at a time. Then uncover the result and explain your first mismatch. Ask why a step is allowed, not only what number it produces.

A pictorial worked case: Hold Out Test Data

Open the final hold-out only after recording the frozen model choice.

The facts we will use

Supplied facts

Frozen plant classifier evaluates12 new grouped leaves: TP 3, FN 1, FP 2, TN 6. Positive means spotted.

Supplied facts

All **input**s and outcomes in this worked example are invented classroom data; no real model run is **claim**ed.

AI ACADEMY / AI-4/23

Frozen final-test results

	predicted spotted	predicted healthy
actual spotted	3	1
actual healthy	2	6

Read a cell using both its row and column labels.

Follow the labelled picture with the matching explanation. Open the vector diagram

AI ACADEMY / AI-4/23

Hold Out Test Data

Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.



1. Record freeze

Record model version and choices before reading these final references.



2. Calculate metrics

Accuracy9/12=75%; precision3/5=60%; recall3/4=75%.



3. Bound claim

Report these twelve held-out cases and their collection conditions; preserve all three error records.

CHANGE ONE THING

The team changes a threshold after inspecting the false negatives.

Explain the changed result using the matching lesson.

Follow the labelled picture with the matching explanation. Open the vector diagram

Read the picture one step at a time

STEP	WHAT HAPPENS AND WHY
1. Record freeze	Record model version and choices before reading these final references.
2. Calculate metrics	Accuracy9/12=75%; precision3/5=60%; recall3/4=75%.
3. Bound claim	Report these twelve held-out cases and their collection conditions; preserve all three error records.

Match each pictured step to the supplied facts. This is a worked teaching case, not an independent assessment.

Why this result follows

Why the result follows

Read the picture in order. Report these twelve held-out cases and their collection conditions; preserve all three error records. The arrows show which recorded **input** or intermediate result each decision uses.

Explain the deciding step aloud before checking the changed case; pointing to the final answer alone does not show how it follows.

What this example does not establish

Keep the stated limit

A hold-out can still be unrepresentative, too small or badly labelled; separation alone does not guarantee reliable estimates.

Keep the stated limit

Read the labelled picture: What is needed for an honest final **claim** about the revised version? Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.

Keep the stated limit

Change one thing: The team changes a threshold after inspecting the false negatives. Predict the result before checking the explanation. What remains the same?

STUDENT LESSON · 7 OF 18

Find and repair a mistake

> Find the unsupported step in the Hold Out Test Data claim and explain a repair using the supplied evidence.

Find the first error

ERROR DETECTIVE

Compare → Diagnose → Repair

Do not patch the answer. Find the earliest unsupported assumption, data problem, or experiment error.

!

COMPARE • DIAGNOSE • REPAIR

Tempting claim: Test data remains unbiased as long as the model never directly fits its labels. Inspect the dataset, split, calculation, error type, subgroup, or experiment record; identify one testable cause; change one factor; rerun the same evidence.

Evidence record	Expected	Actual	Likely cause / next test
normal case	passes		
boundary case	defined behavior		
missing case	safe fallback		
shifted case	measured limit		

CLAIM TO CHALLENGE Test data remains unbiased as long as the model never directly fits its labels.

Diagnose one cause and name the cleanest next experiment:

A closer explanation

A report can honestly say 15 of 20 correct on this fictional held-out set. It cannot say the model is 75 percent reliable for every child, school or future **input**. Do not fabricate more examples to make a result sound decisive.

> **One planned case returns no prediction. What must the report retain?**

STUDENT LESSON · 8 OF 18

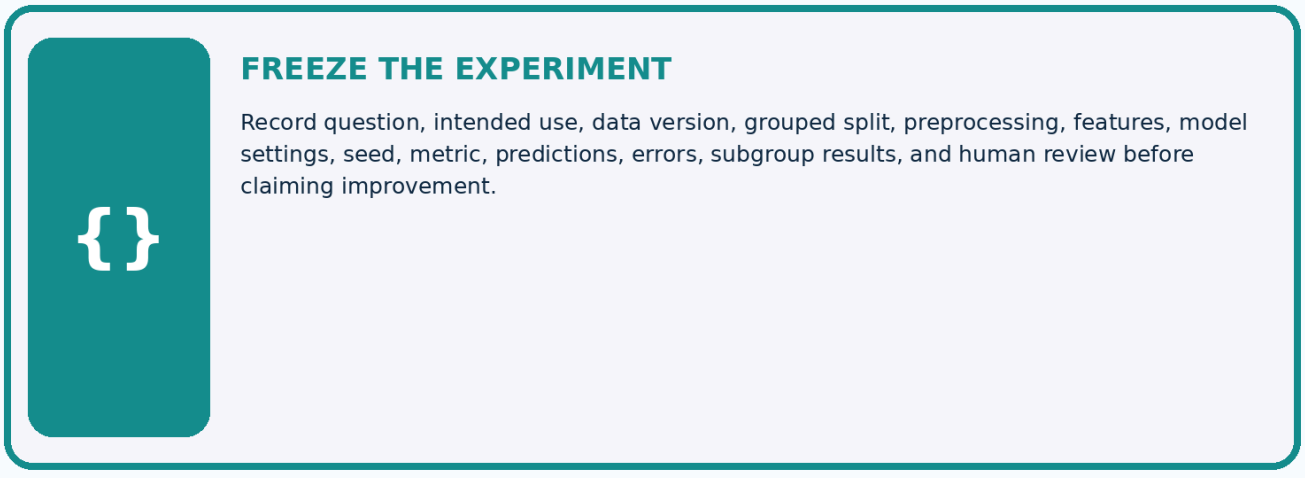
Practise together

> Practise Hold Out Test Data with support, then explain the deciding step in your own words.

ML STUDIO

Specify Before You Run

A trustworthy result can be reproduced from the recorded data, code, settings, and evaluation rule.



Experiment record

> **Question to explain**
>
> QUESTION + intended use + success threshold

DATA version + grouped split + leakage check

CODE/pre**process**ing + features + model + settings + seed

METRICS + positive class + units + denominators

PREDICTIONS + errors + subgroup slices + limitations

HUMAN reviewer + release, revise, limit, or stop decision

Write the minimum record needed to reproduce today's result:

A closer explanation

After seeing three missed blocked cases, the team changes its threshold. The old test has now informed development. Preserve the first result and label later scores as post-test development.

A new independent test is needed for a new untouched-test **claim**; repeatedly splitting the same inspected rows does not erase what was learned.

Practise with less help: Compare a baseline

Use workbook W2 for the connected example “Final testing and honest **claim**s”. First fill the **input** and rule boxes. Try the middle steps before asking for a hint. After a hint, try the next step yourself.

Plan	Do	Check
What is given? Which rule or procedure applies?	Record each intermediate step.	Does the result follow? What remains unknown?

Explain your trace to a partner. Your partner points to one step to check. Swap roles on the next case. Each person writes their own explanation.

Check your reasoning

Use the worked case for Final testing and honest **claim**s. Proposed explanation: “A final answer is enough”. Decide whether this explanation is supported.

Point to the step or supplied fact that decides; explain your repair if needed.

Answer on your own before discussion. If you are unsure, say which step you need help with.

Explain the picture

What is needed for an honest final **claim** about the revised version? Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.

> **My explanation using a pictured step and the given facts**
>
> _____
>
> _____
>
> _____
>
> _____

This is supported practice. Keep the separately named independent case for an attempt before feedback.

STUDENT LESSON · 9 OF 18

Run the investigation

> Investigate Hold Out Test Data. Record the prediction, actual observation or labelled paper trace, explanation and next test separately.

EVIDENCE LAB

Predict → Run → Calculate → Interpret

Keep the test conditions fixed, record every result, and change only one planned factor.

LAB

PREDICT → RUN → CALCULATE → INTERPRET

Split-Integrity Court: classify workflow actions as allowed or contaminating, then redesign the evidence trail.

1. Write the hypothesis and expected **evidence** before running.

2. Split-Integrity Court: classify workflow actions as allowed or contaminating, then redesign the **evidence** trail.

3. Run one normal, one boundary, one missing, and one shifted case.

4. Calculate the metric from raw counts or row-level errors.

5. Interpret what changed, what did not, and what remains unsafe or unknown.

Case · Prediction · Actual · Calculation / evidence · Meaning / next test

> **Case:** Normal

Prediction: _____

Actual: _____

Calculation / evidence: _____

Meaning / next test: _____

Case: Boundary

Prediction: _____

Actual: _____

Calculation / evidence: _____

Meaning / next test: _____

Case: Missing

Prediction: _____

Actual: _____

Calculation / evidence: _____

Meaning / next test: _____

Case: Shifted

Prediction: _____

Actual: _____

Calculation / evidence: _____

Meaning / next test: _____

Plan → try → check

Before trying: what do I expect, and why? While trying: which step could first go wrong? Afterwards: what did I actually observe, and what should change? Keep “not run” if you only traced the procedure.

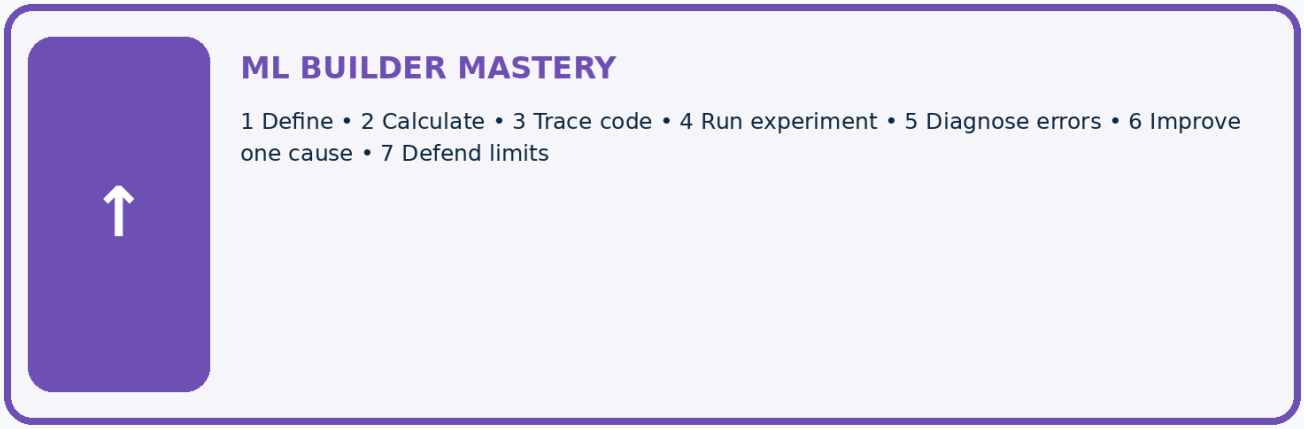
Build your understanding

> Move from recall to tracing, diagnosis and transfer in Hold Out Test Data. Explain why each step is justified.

MASTERY

From Knowing the Word to Owning the Idea

Move upward only when you can produce **evidence** without copying the example.



DEFINE Use all four terms precisely: **holdout**, **contamination**, **model selection**, **final estimate**.

CALCULATE Reproduce the worked **evidence** with units and denominator.

TRACE Explain the data, code or decision path.

DIAGNOSE Test data remains unbiased as long as the model never directly fits its labels.

TRANSFER Solve a fresh dataset or model case.

IMPROVE Change one cause and rerun fixed **evidence**.

DEFEND State intended use, metric, limitation, and human responsibility.

MASTERY RULE A coached calculation becomes mastery only after a different case is completed independently.

STUDENT LESSON · 11 OF 18

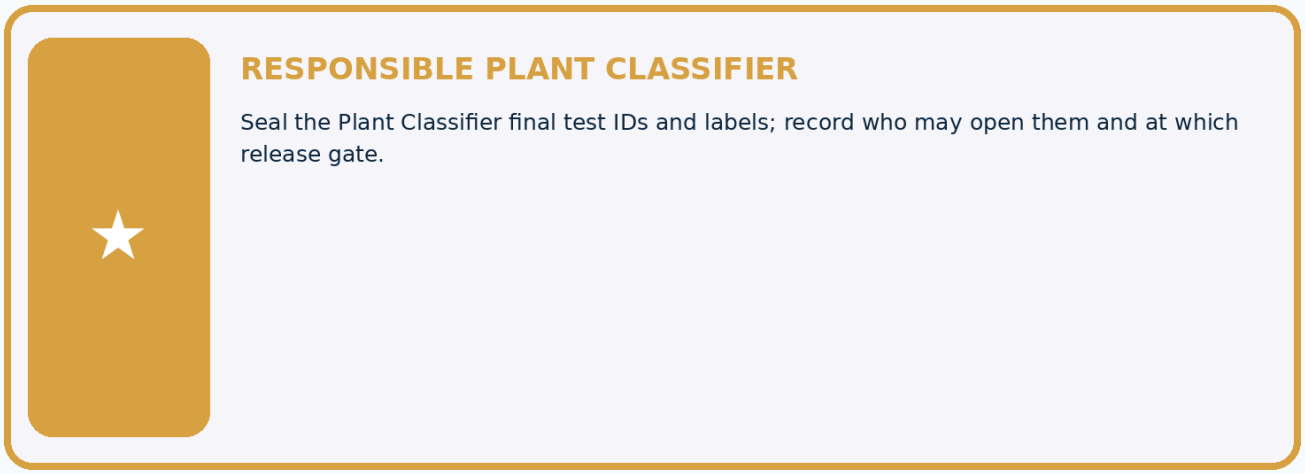
Connect your project

> Apply this week's idea to Responsible Plant Classifier. Keep an annotated decision and the evidence supporting it.

CUMULATIVE PROJECT

Responsible Plant Classifier

Every week adds one reviewable artifact to the same **evidence**-based capstone.



THIS WEEK'S CHECKPOINT Seal the Plant Classifier final test IDs and labels; record who may open them and at which release gate.

Which requirement or risk does this artifact address?

Which dataset, split, code, and metric version does it use?

What normal and shifted cases were tested?

What remains uncertain or weak?

What decision still belongs to a teacher or qualified human?

RESPONSIBLE RELEASE Never use this classroom prototype for medical, pesticide, agricultural, or other consequential decisions.

My **evidence**-linked project decision:

Evidence for your project

For your Responsible Plant Classifier, save the **input**, procedure, observed result and limitation of this check.

Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding.

STUDENT LESSON · 12 OF 18

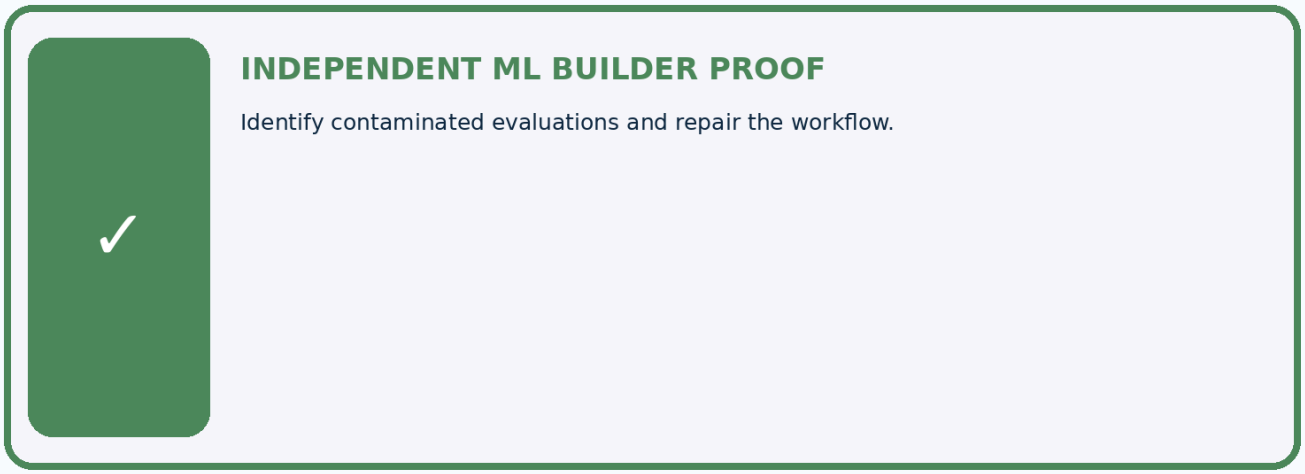
Show what you know

> Use a genuinely new case for independent evidence. Previously discussed examples remain practice; use the separately named independent case before feedback.

INDEPENDENT PROOF

Explain • Calculate • Transfer • Defend

Complete this with a fresh case after guided practice and compare it with the mastery criteria.



What success looks like

REQUIRED PROOF Identify contaminated evaluations and repair the workflow.

- ☐ Define the task, feature/target roles, and intended-use boundary.
- ☐ Show the calculation, code trace, or experiment record.
- ☐ Test a normal case and one revealing boundary, missing, or shifted case.
- ☐ Diagnose one failure without changing several factors at once.
- ☐ Name a limitation, fairness concern, and the human-review action.

Independent **evidence** and defense:

My level: ☐ Not yet ☐ With one hint ☐ Independently ☐ I can teach and defend it

Connect to the next lesson

NEXT BRIDGE Carry the experiment **evidence** into: Train, Validation, Test.

STUDENT LESSON · 13 OF 18

Study a complete case

> Revisit the complete Hold Out Test Data case for supported practice. Repeated examples do not become fresh assessments.

Week 23 Worked case

Use the supplied details to practise the weekly concept before attempting the independent question. This case extends the illustrated lesson.

The situation

A final-test folder has twenty leaves; its labels are opened weekly to choose models.

The question

> Is it still untouched?

Worked explanation

No. It functions as validation. Reserve a genuinely unused set for final evaluation or disclose the limitation.

Explain it in your own words

> Which detail supports the answer, and why does it matter?

Write or draw your reasoning and evidence.

Trace the supplied case visually

A repeatedly inspected test set is no longer final

Evidence or stage	Value or result
Reserved folder	Twenty leaves
Weekly inspection	Labels influence model choice
Actual role	Validation
Repair	Reserve genuinely unused final cases

A final-test folder has twenty leaves; its labels are opened weekly to choose models.

Is it still untouched?

No. It functions as validation. Reserve a genuinely unused set for final evaluation or disclose the limitation.

STUDENT LESSON · 14 OF 18

Try a new case independently

Attempt the reserved new case before feedback. Record support; the earlier examples below remain additional practice.

Independent case: Apply independently

A new frozen test has TN 7 FP 1 FN 2 TP 2. Calculate accuracy, precision and recall and state a bounded **claim**.

My independent explanation, trace and limits

Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

Week 23 Independent application

Close the worked explanation. Apply the idea to this different question without copying.

You inspect final test failures, change features to fix them and report the new score on that same test set as untouched. What is wrong?

My answer, with a trace, calculation or supporting **evidence**

Write or draw your reasoning and evidence.

Create and test

Change one relevant detail. Predict the result before testing. Include a boundary or missing-information case when it helps reveal a limit.

My changed **input** and expected result

Write or draw your reasoning and evidence.

Connect to your project

Seal the Plant Classifier final test IDs and labels; record who may open them and at which release gate.

The **evidence** I will keep

Write or draw your reasoning and evidence.

Your teacher has the independent answer and scoring criteria. Complete the matching workbook week to keep your practice **evidence** together.

Transfer practice from the original programme

After seeing test errors, a team adjusts its threshold and reports the improved score on that same test. Rename the set's role and describe an honest final report if no new data exists.

Explain the mechanism, evidence and one limit.

Use feedback to improve

Attempt the independent case before hints. Record whether you worked alone, used a hint or followed a model. After feedback, fix the first unsupported step and explain why it changed.

A later check uses a changed case, not a copied answer.

STUDENT LESSON · 15 OF 18

Your lesson route

> Use this map to navigate Hold Out Test Data, connect the goals to tasks and keep one evidence record for each attempt.

Start with this question: Open the final hold-out only after recording the frozen model choice.

By the end, I can	How I will show it
Freeze the full evaluation procedure before testing	Explain the deciding step, show a trace or test, and state one limit.
Report a held-out result with counts and limitations	Explain the deciding step, show a trace or test, and state one limit.
Recognize when test feedback has become development	Explain the deciding step, show a trace or test, and state one limit.

Remember → watch and explain → try with help → investigate → try independently → keep project **evidence**. Your teacher may spread this lesson across several classes.

Keep your first **prediction**. Compare it with what the **evidence** shows and explain any change.

STUDENT LESSON · 16 OF 18

Finish and return

> Explain Hold Out Test Data without notes, identify a remaining question and plan a later check with a changed case.

Close your notes. Explain this goal in your own words: **Freeze** the full evaluation procedure before testing

> **My deciding step, evidence and one remaining question**
>
> _____
>
> _____
>
> _____
>
> _____

If you are unsure, return to the worked picture and explain one arrow. If you are ready, change an **input** or assumption and justify your **prediction**.

Save your workbook **evidence**. At the next review, try a different case before opening your old answer.

STUDENT LESSON · 17 OF 18

Supported practice, repair and portfolio

> Attempt supported questions before using hints or feedback. Review a missing building block, then keep the completed work.

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

Open the final hold-out only after recording the frozen model choice.

Practice 1: explain the deciding evidence

Which labelled step matches this detail? Report these twelve held-out cases and their collection conditions; preserve all three error records.

1. Record **freeze**

2. Bound **claim**

3. Calculate metrics

My selected conclusion, deciding fact and explanation

Hint · try first

Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

Feedback · compare after trying

After your attempt, compare with the supported explanation: Bound **claim** Report these twelve held-out cases and their collection conditions; preserve all three error records. Read the picture in order.

Report these twelve held-out cases and their collection conditions; preserve all three error records. The arrows show which recorded **input** or intermediate result each decision uses.

Explain the deciding step aloud before checking the changed case; pointing to the final answer alone does not show how it follows.

Practice 2: explain the deciding evidence

Changed case · predict first

Changed-case practice: The team changes a threshold after inspecting the false negatives. Which conclusion is supported by the supplied facts?

1. We can decide the result without examining the changed **input** or the procedure.

2. This set is now development **evidence** for the revised model.

3. This one example guarantees correct results for every future **input**.

My selected conclusion, deciding fact and explanation

Hint · try first

Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.

Feedback · compare after trying

After your attempt, compare with the supported explanation: This set is now development **evidence** for the revised model. Using final outcomes to change the model breaks the untouched hold-out role.

Repair a missing building block

Review Train a Regression Model (ai-4/22). Goal: Fit a line from a declared **training** candidate set. Short practice: May day3's evaluation label choose the candidate line?

Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.. Return to this week and explain what you changed.

Review Mean, Median, and Baselines (ai-4/20). Goal: Fit mean and median constant baselines on **training** targets. Short practice: Does either original baseline win under evaluation MAE?

Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.. Return to this week and explain what you changed.

Review Precision and Recall (ai-4/17). Goal: Choose the denominator that answers the question. Short practice: Were the original precision and recall denominators the same?

Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.. Return to this week and explain what you changed.

Keep your completed work

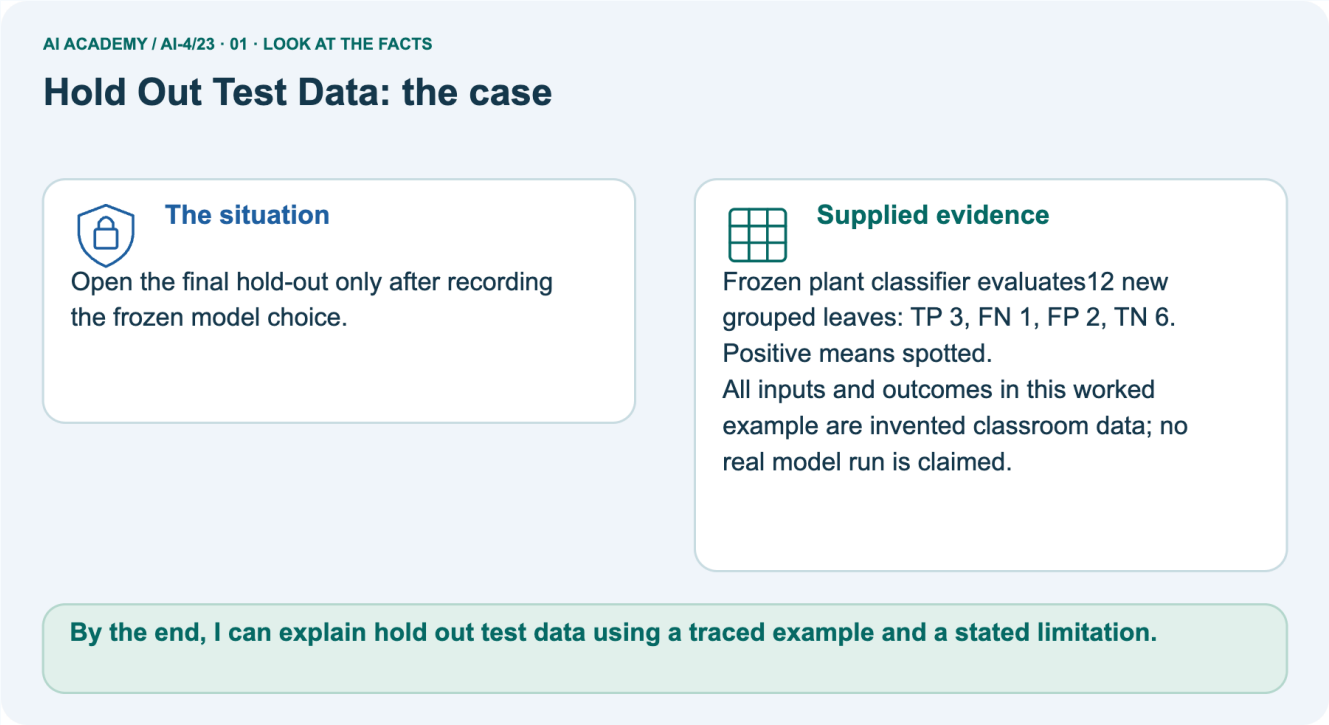
Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run **evidence**. Open the HTML copy to read or print it.

Drafts stay in this browser when storage is available; clear drafts on a shared device.

Visual concept journey

> Explain each Hold Out Test Data graphic using its labels, deciding evidence and relevant limitation.

01. Hold Out Test Data: the case



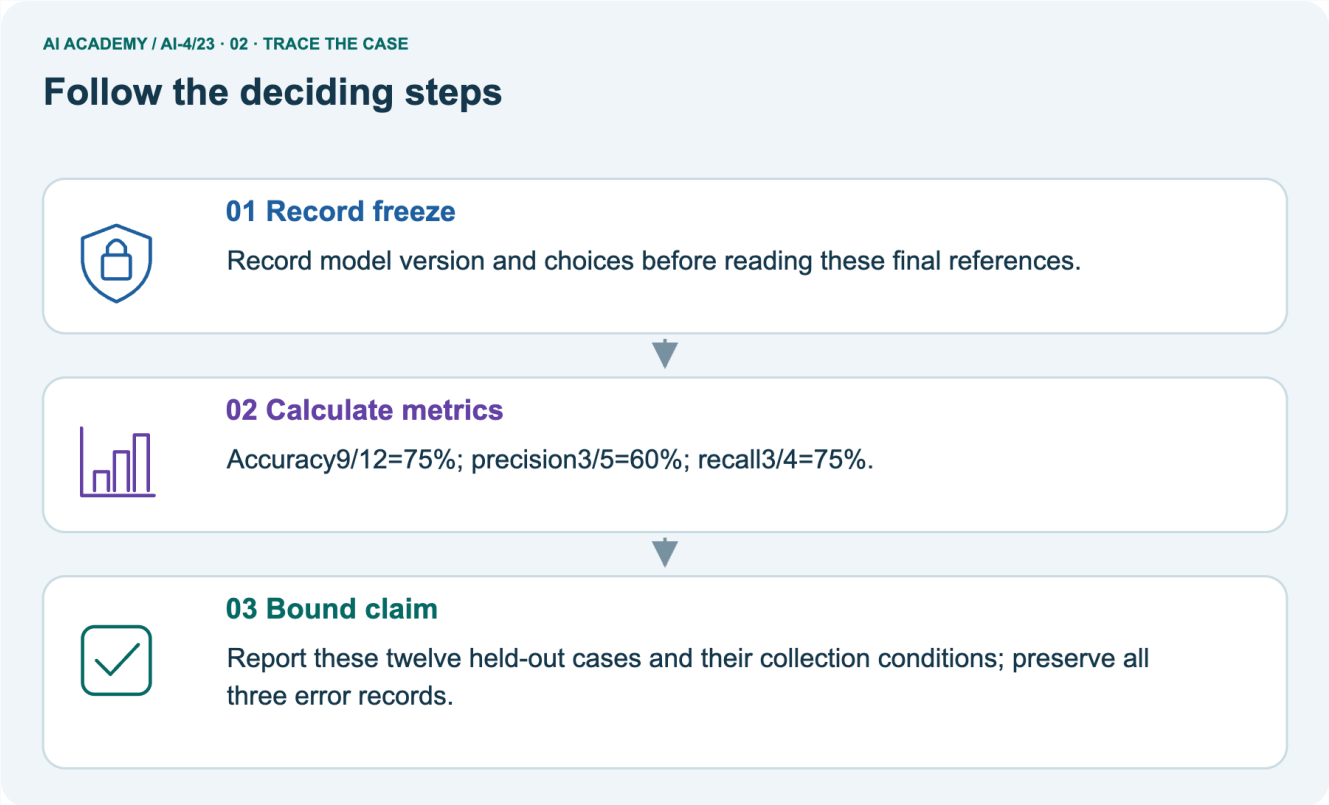
Follow the labelled picture with the matching explanation. Open the vector diagram

Open the final hold-out only after recording the frozen model choice. Supplied facts: Frozen plant classifier evaluates12 new grouped leaves: TP 3, FN 1, FP 2, TN 6. Positive means spotted.

All **input**s and outcomes in this worked example are invented classroom data; no real model run is **claim**ed.

> Point to the given fact that matters most. Explain why it matters.

02. Follow the deciding steps

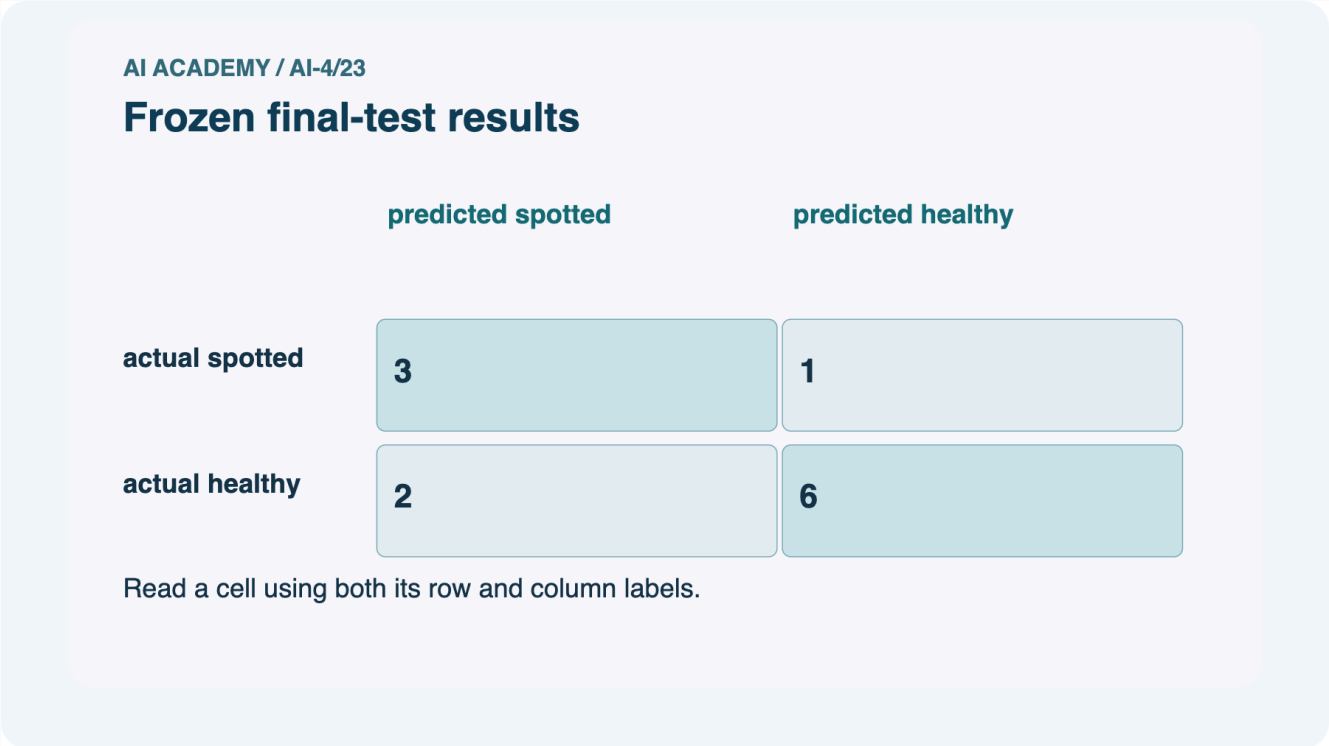


Follow the labelled picture with the matching explanation. Open the vector diagram

Record **freeze**: Record model version and choices before reading these final references. → Calculate metrics: Accuracy9/12=75%; precision3/5=**60%**; recall3/4=75%. → Bound **claim**: Report these twelve held-out cases and their collection conditions; preserve all three error records.

> Point to each numbered stage and explain which recorded **input** it uses.

03. Frozen final-test results

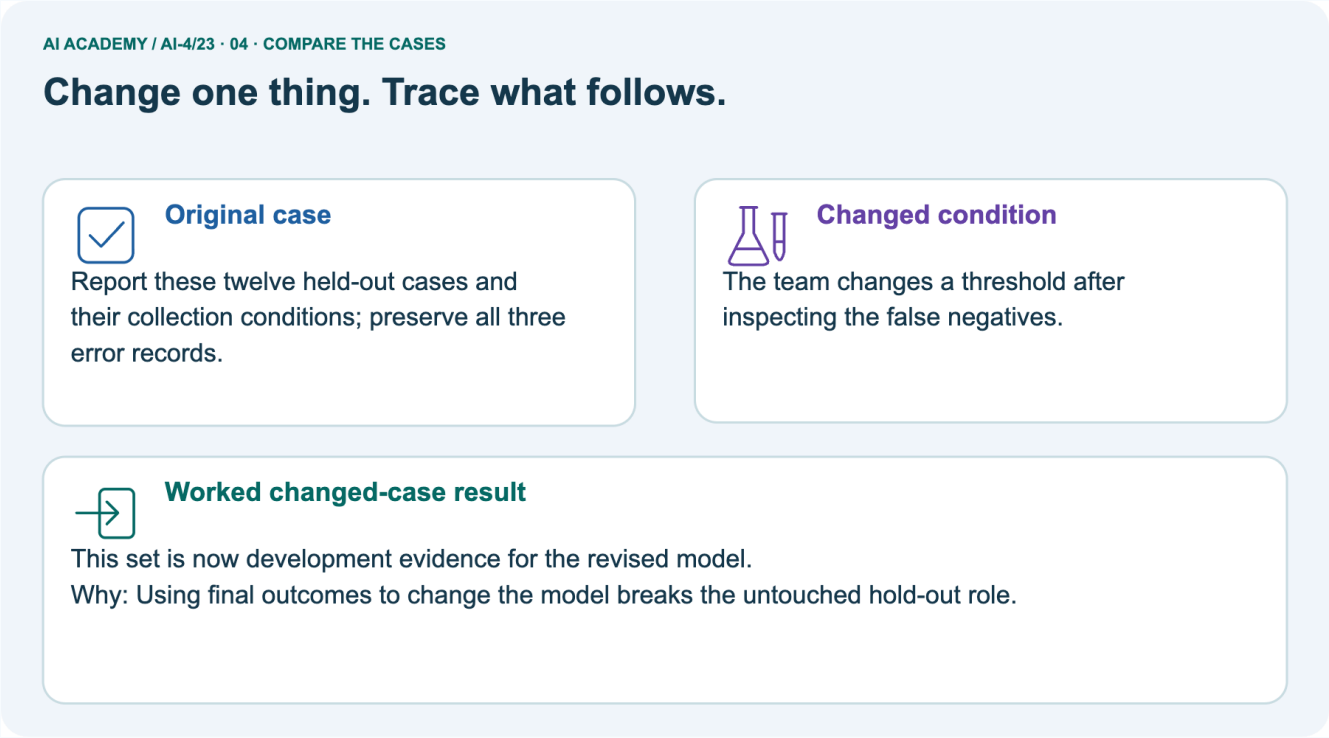


Follow the labelled picture with the matching explanation. Open the vector diagram

Frozen final-test results. actual spotted: predicted spotted = 3, predicted healthy = 1. actual healthy: predicted spotted = 2, predicted healthy = 6.

> Read the labels and values. Explain the relationship the model shows.

04. Change one thing. Trace what follows.



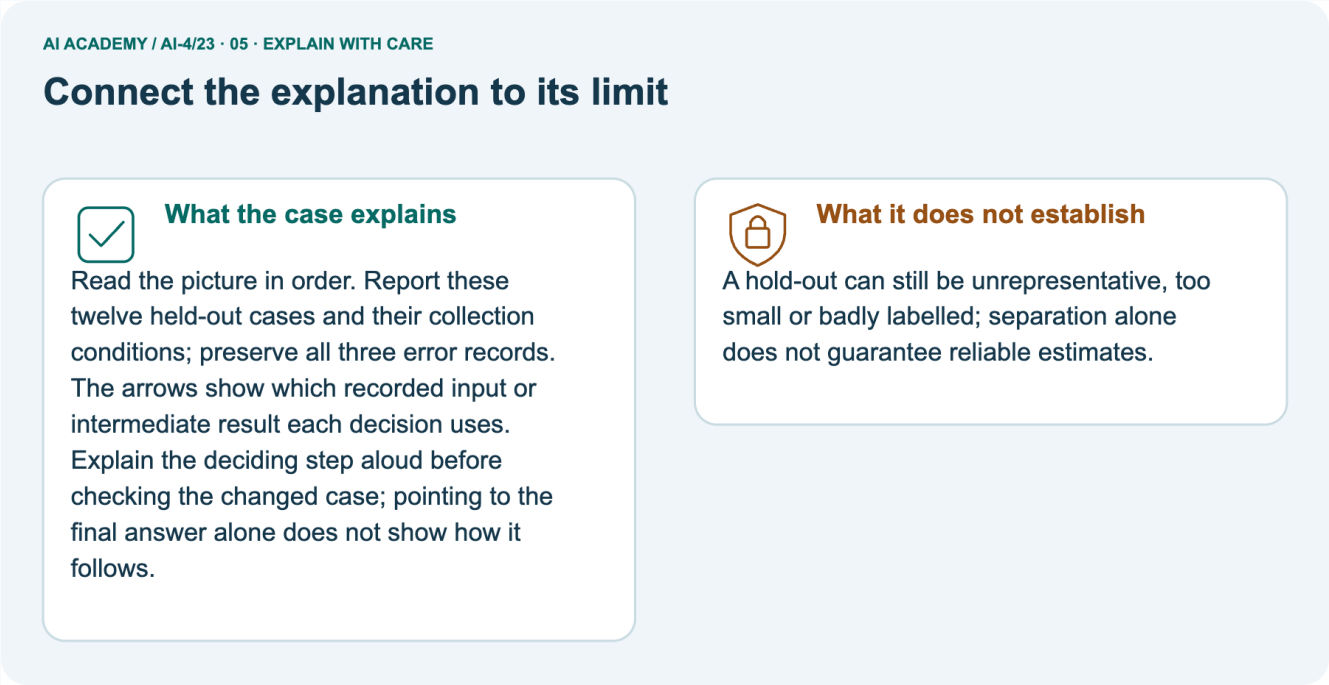
Follow the labelled picture with the matching explanation. Open the vector diagram

Original case: Report these twelve held-out cases and their collection conditions; preserve all three error records. Changed condition: The team changes a threshold after inspecting the false negatives.

Expected result: This set is now development **evidence** for the revised model. Reason: Using final outcomes to change the model breaks the untouched hold-out role.

> Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

05. Connect the explanation to its limit



Follow the labelled picture with the matching explanation. Open the vector diagram

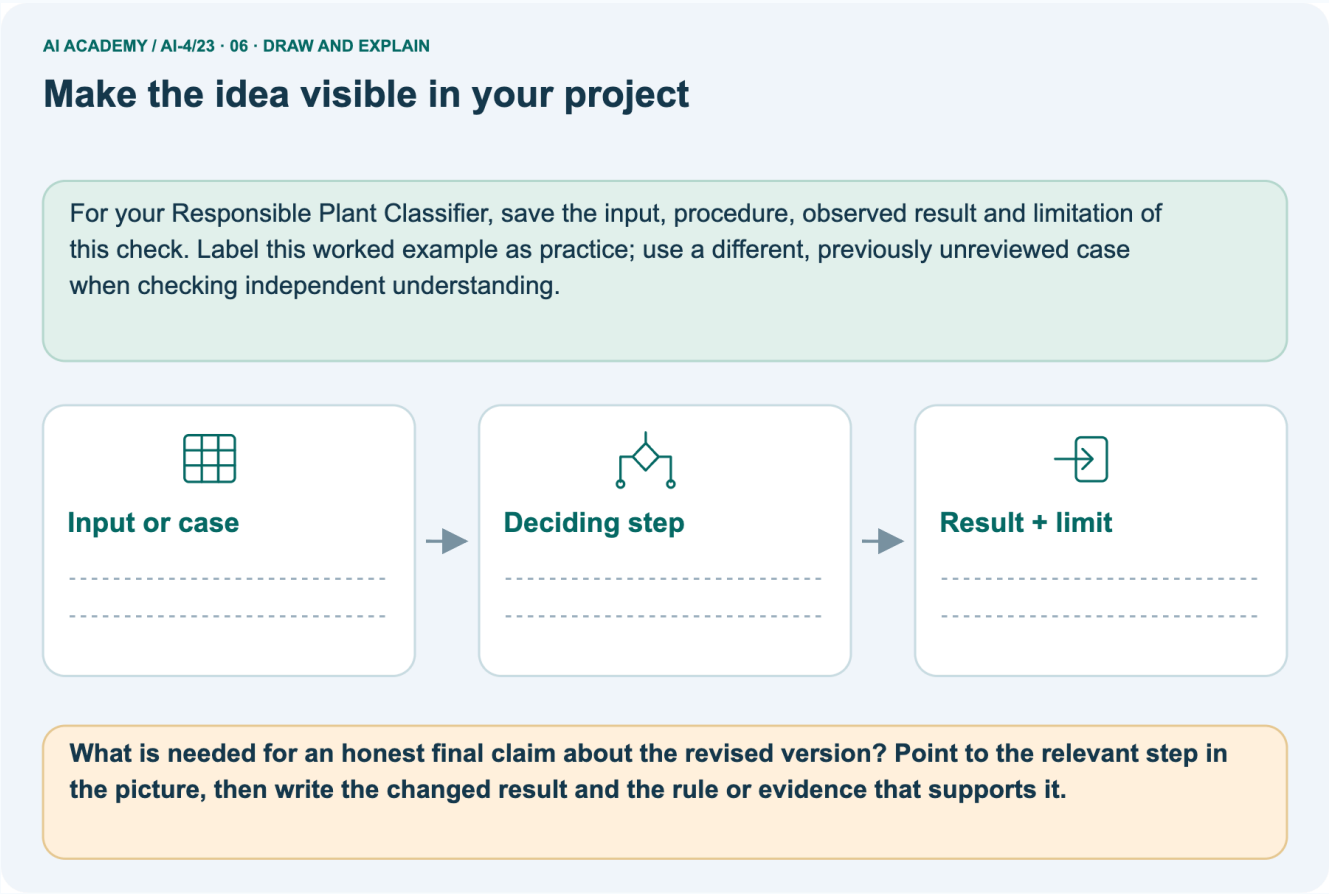
Read the picture in order. Report these twelve held-out cases and their collection conditions; preserve all three error records. The arrows show which recorded **input** or intermediate result each decision uses.

Explain the deciding step aloud before checking the changed case; pointing to the final answer alone does not show how it follows.

Limit: A hold-out can still be unrepresentative, too small or badly labelled; separation alone does not guarantee reliable estimates.

> Explain the case, then point to a limit on the **claim**.

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

For your Responsible Plant Classifier, save the **input**, procedure, observed result and limitation of this check.

Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding. Draw your **input**, deciding step, result and limit.

Supported question: What is needed for an honest final **claim** about the revised version? Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.

> Draw your project trace and justify the deciding step using **evidence**.

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.